Hackathon Finder

Product Requirements Document (PRD)

Version 1.0 | 14 Oct 2026 | Status: draft | Working name: HackRadar

1. Summary

A web app (installable as a PWA) that gathers hackathons from many platforms and colleges into one searchable, filterable list, highlights what is closing soon, and sends alerts so students never miss a registration deadline.

2. Problem statement

- Hackathons are spread across many sites (Unstop, Devfolio, Devpost, MLH, Hack2skill, college pages and social posts).
- Registration windows are short and easy to miss; many events open and close quietly.
- College and local hackathons often appear on no major platform at all.
- Existing sites filter poorly for what a student cares about: eligibility, city, team size, prize and theme together.

3. Vision, goals and non-goals

Vision: the one place a student checks to know which hackathons they can join right now.

Goals (V1)	Non-goals (V1)
Aggregate 3 to 4 sources automatically	Handling registrations or payments inside the app
Fast search and rich filters	Team formation and chat
Deadline-first discovery and alerts	Native mobile apps (PWA only)
Crowdsourced listings with moderation	Global coverage from day one (start India-first)
Run entirely on free tiers	Monetisation

4. Target users

Persona	Needs	Pain today
Student hacker (2nd to 4th year, tier-2 or tier-3 city)	Find hackathons they are eligible for, near them or online, before deadlines	Checks 5 sites manually, misses deadlines
Beginner	Beginner-friendly, low-pressure events with small team sizes	Cannot tell which events suit a first-timer
Organiser (college club)	Get their event seen by students	No reach beyond their own college

5. User stories

ID	As a...	I want to...	So that...	Priority
US-1	Student	search by keyword	I find relevant hackathons fast	P0
US-2	Student	filter by mode, city, prize, team size, theme and eligibility	I only see events I can join	P0
US-3	Student	sort by closing soon	I never miss a deadline	P0
US-4	Student	open a detail card with a clear summary and register link	I decide quickly and go register	P0
US-5	Student	bookmark hackathons	I can track the ones I like	P1
US-6	Student	add events to my calendar	deadlines show up in Google Calendar	P1
US-7	Student	get Telegram or email alerts for matching events	I hear about them without checking	P1
US-8	Organiser	submit my hackathon	students discover it	P1
US-9	Student	see a match score for each event	I focus on best-fit hackathons	P2
US-10	Student	see an AI summary of each event	I skip reading long pages	P2

6. Feature list and priorities

Feature	Description	Priority
Multi-source aggregation	Scheduled ingestion from Unstop, Devfolio and 1 to 2 more, normalised and deduplicated	P0
Search and filters	Keyword plus mode, city, prize, team size, theme, student-eligible, status	P0
Deadline radar	Closing-soon sort, live countdowns, 48-hour highlight	P0
Detail card	Dates, prize, eligibility, summary, link to original	P0
Accounts and bookmarks	Sign in with email or Google; save events	P1
Calendar feed	Private .ics link that stays in sync	P1
Alerts	Telegram bot and email for new matches and closing soon	P1
Submit a hackathon	Form plus admin moderation queue	P1
Match score	Rank by skills, eligibility, urgency and prize	P2
AI summaries	3-line summary and auto tags	P2
Effort vs reward score	Combined worth-it indicator	P2
Team finder	Post needs and find teammates	Later

7. Functional requirements

- **FR-1** The system shall refresh listings from every configured source at least every 6 hours.
- **FR-2** Each listing shall store title, organiser, mode, location, event dates, registration deadline, prize, team size, eligibility, tags and source link.
- **FR-3** Duplicate listings from different sources shall appear once, with links to all sources.
- **FR-4** Users shall filter by any combination of mode, city, status, tag, prize minimum and team size.
- **FR-5** Status (open, closing soon, closed) shall be computed from dates, with closing soon meaning within 72 hours.
- **FR-6** Closed hackathons shall be hidden from the default view but remain reachable by link.

- **FR-7** Signed-in users shall be able to bookmark, manage alert rules and get a private calendar link.
- **FR-8** The system shall never send the same alert to the same user twice.
- **FR-9** Submitted hackathons shall stay unpublished until an admin approves them.
- **FR-10** Every card shall link to the original source page for registration.

8. Non-functional requirements

Area	Requirement
Performance	List page loads in under 2 seconds on a typical 4G connection; filter responses under 500 ms
Freshness	Data at most 6 hours old; deadlines checked hourly for alerts
Cost	Rs 0 hosting; stay inside free-tier limits
Reliability	One failing source must not affect others; best-effort uptime is acceptable for V1
Usability	Mobile-first, installable PWA, readable without sign-in
Security and privacy	RLS on user data, minimal personal data, secrets never in the repo
Accessibility	Keyboard navigable, sufficient colour contrast, semantic HTML

9. Success metrics

Initial targets to review after the first month of real use:

Metric	Why it matters	Initial target
Hackathons indexed and live	Coverage	150+ active listings
Sources healthy	Reliability	At least 90% of scheduled runs succeed
Data freshness	Trust	95% of listings updated within 12 hours
Weekly active users	Adoption	100+ within a month (campus word of mouth)
Alert click-through	Value of alerts	30%+
User submissions approved	Crowdsourcing edge	20+ college events

10. Release plan

Week	Deliverable
1	Database schema, 2 scrapers (Unstop, Devfolio), basic list page
2	Search, filters, deadline radar, detail card
3	Accounts, bookmarks, calendar feed, submit form with moderation
4	Telegram and email alerts, AI summaries, polish and launch to campus groups

11. Risks and assumptions

- Sources may restrict scraping or change formats; mitigated by adapters, multiple sources and user submissions.
- Free-tier limits could throttle growth; mitigated by caching and monitoring.
- Assumes students prefer a single curated feed over checking each platform.
- Assumes enough public data exists per listing to power filters (prize and team size are sometimes missing).

12. Open questions

- India-first or global at launch? (Recommended: India-first with 3 to 4 sources.)

- Should browsing require login, or only bookmarks and alerts? (Recommended: browsing is open.)
- Final product name and domain.
- Who moderates submissions at the start, and how fast must they be reviewed?